

Problem 1:*The Meaning of Life*

In 2021, Pew Research Center conducted a survey of 2,596 adults in the United States (report [here](#)). They asked "What aspects of your life do you currently find meaningful, fulfilling, or satisfying?" They did not report raw data, but below is a generated dataset based on reported results.

Topic	Frequency
Family	716
Friends	300
Material Well-being	265
Career	262
Challenges	285
Spirituality	250
Society	210
Health	172
Hobbies	136

(a)

Compute the Relative Frequencies for each response category.

(b)

Construct a bar graph of the Relative Frequencies.

(c)

Reporting on these poll results suggested that over half of respondents chose one of the top three topics. Does the data support this conclusion?

Problem 2:*Board Game Weights*

The file `bgg2022.csv` on Canvas contains a dataset of every board game on the popular site “Board Game Geek” as of 2022. These data can be used to answer these questions. Note the following information about the variables:

- The column `AveRating` gives the average user review score of the game, where 1 is “Awful — defies game description” and 10 is “Outstanding — will always enjoy playing.”
- The column `AveWeight` gives the average user assessment of the “weight” of each game. Weight refers to the overall complexity of the game, not to any weighted average coefficient or any such thing. Weights 1 correspond to “Light” and weights of 5 correspond to “Heavy.”

(a)

Create a single box plot for the review scores (`AveRating`) for all games in this dataset. Remember to add a descriptive title and axis labels.

(b)

Based on the box plot, is the distribution of all game review scores skewed? If so, in what direction?

(c)

Create Box Plots for the average weight of games by whether it is ranked as a Family Game or not. If `Family Game Rank` is blank (coded as `NaN`), then it is not ranked as a Family Game. I recommend adding a new column using the function `np.isnan`.

Games with an `AveWeight` value of 0 have not had their weight rated by any users. These should be removed and ignored for **all** of this analysis.

(d)

Based on the box plots above, approximately what percentage of Non-Family games are heavier than 75% of Family games?

Problem 3:*Baseball Hall-of-Famers*

Baseball Hall-of-Famers (HoF) played during different eras of baseball. One common classification of eras (see [here](#)) is the following:

Era	Years
19th Century	– 1900
Dead Ball	1901 – 1919
Live Ball	1920 – 1941
Integration	1942 – 1960
Expansion	1961 – 1976
Free Agency	1977 – 1993
Steroid Era	1994 – 2005
Post-Steroid Era	2006 –

For this exercise, define the era of a player based on the mid-point of their career (**rounding up if the mid-point falls between eras**).

Using the file `hofbatting.csv`, containing non-pitching HoFs as of 2024, classify each player according to their era to answer the following questions.

(a)

Create a Bar Graph for the number of HoFs from each era as of 2024.

(b)

One era of baseball should stand out as having produced far more Hall-of-Famers than the rest. Which era is it?

(c)

Create a histogram showing the distribution of non-pitching HoFs' Mid-Career year. Set bins by decade, i.e., $[1880, 1890)$, $[1890, 1900)$, $\dots$, $[2010, 2020)$.

(d)

There are two major dimensions to hitting: the ability to get on base (measured by the on-base percentage **OBP**) and the ability to advance runners already on base (measured by the slugging percentage **SLG**). Create a scatterplot of OBP (x-axis) vs. SLG (y-axis).

(e)

In the scatterplot above, are there any outliers? If so, identify them by name.

Don't be concerned about any specific "rule" for finding outliers; just take notice of any points that seem unusual or fall far from the center of mass for this bivariate distribution.

Specifically, what is the name of the individual represented by the point in the bottom-right with lower SLG than would be expected given his OBP (or conversely higher OBP than would be expected given his SLG)?

(f)

Consider a combined metric for hitting, the On-base Plus Slugging (OPS) statistic, which is the sum of OBP and SLG. Standardize this data (i.e., calculate the z-scores), then create a scatterplot with OPS on the y-axis and Mid-Career Year on the x-axis.

(g)

A common rule of thumb for identifying possible outliers is an absolute standardize value greater than 3. Which Hall-of-Famers fall into this category, showing unusually high batting prowess as measured by OPS?